Per-Asset Independent Portfolios

Three $10,000 sub-portfolios — each asset runs its own winner strategy

Backtest 2018-01-01 -> 2026-04-01 | Binance spot | 0.1% fees + 5-tick slip

	Start	Final	Multi	CAGR	Sharpe	MaxDD	Calmar	Trades	Win %
BTCUSDT V4C_range_kalman @ 4h	**$10,000**	**$156,611**	**15.7x**	**+39.6%**	**1.32**	**-28.8%**	**1.37**	**78**	**42.3%**
ETHUSDT V3B_adx_gated @ 4h	**$10,000**	**$395,503**	**39.6x**	**+56.2%**	**1.26**	**-33.8%**	**1.66**	**88**	**43.2%**
SOLUSDT V2B_volume_breakout @ 4h	**$10,000**	**$566,967**	**56.7x**	**+104.7%**	**1.35**	**-51.5%**	**2.03**	**94**	**33.0%**

COMBINED 3 x $10,000 sub-portfolios = 30,000

Metric	Strategies	Buy & Hold
Final equity	**$1,119,079**	$364,283
Total return	**+3630.3%**	+1114.3%
CAGR	**+55.1%**	+35.4%
Sharpe	**1.43**	0.78
Max DD	**-32.2%**	-94.1%
Calmar	**1.71**	0.38

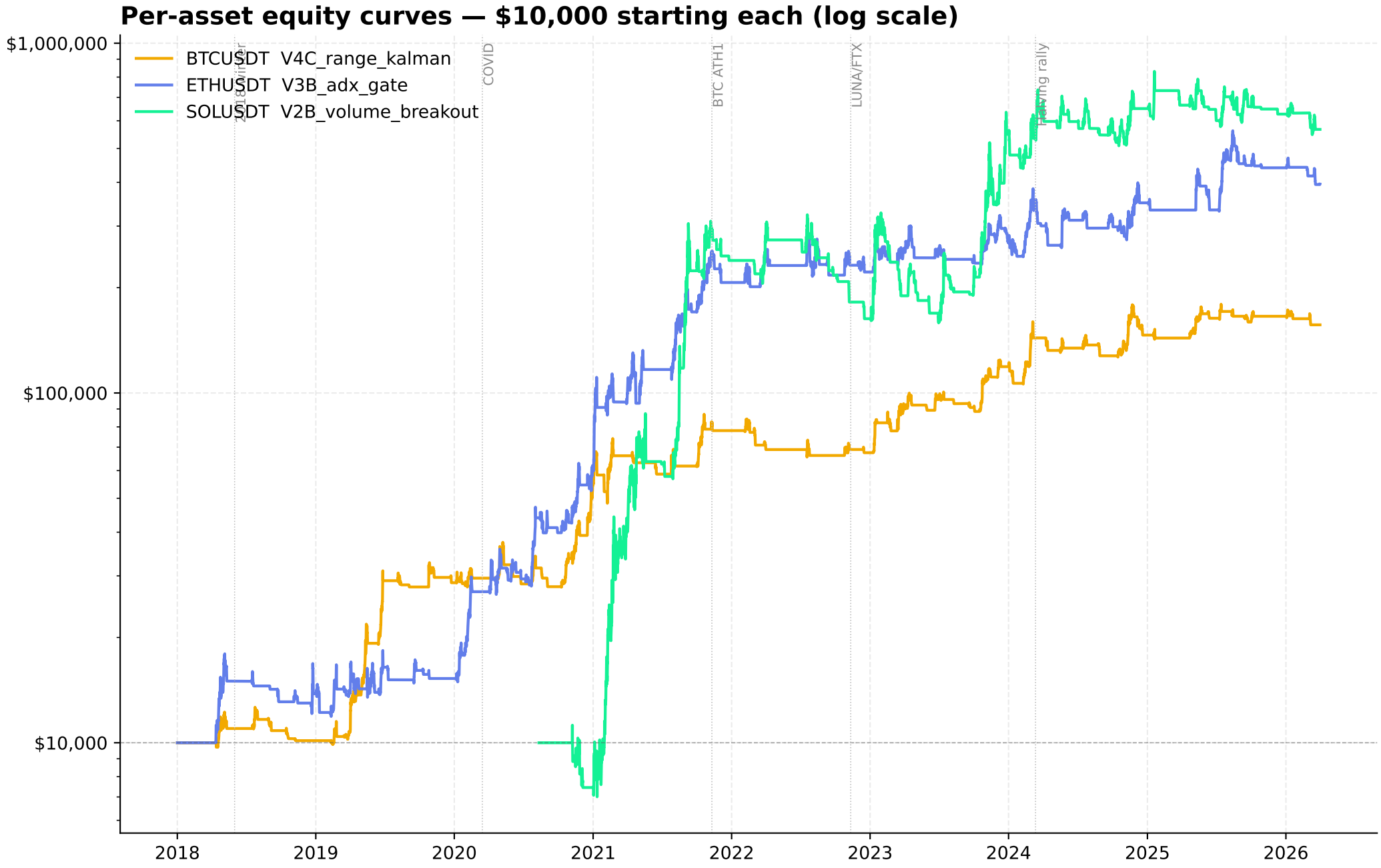
Equity ratio: 3.07x | DD ratio: 0.34x | Calmar ratio: 4.55x

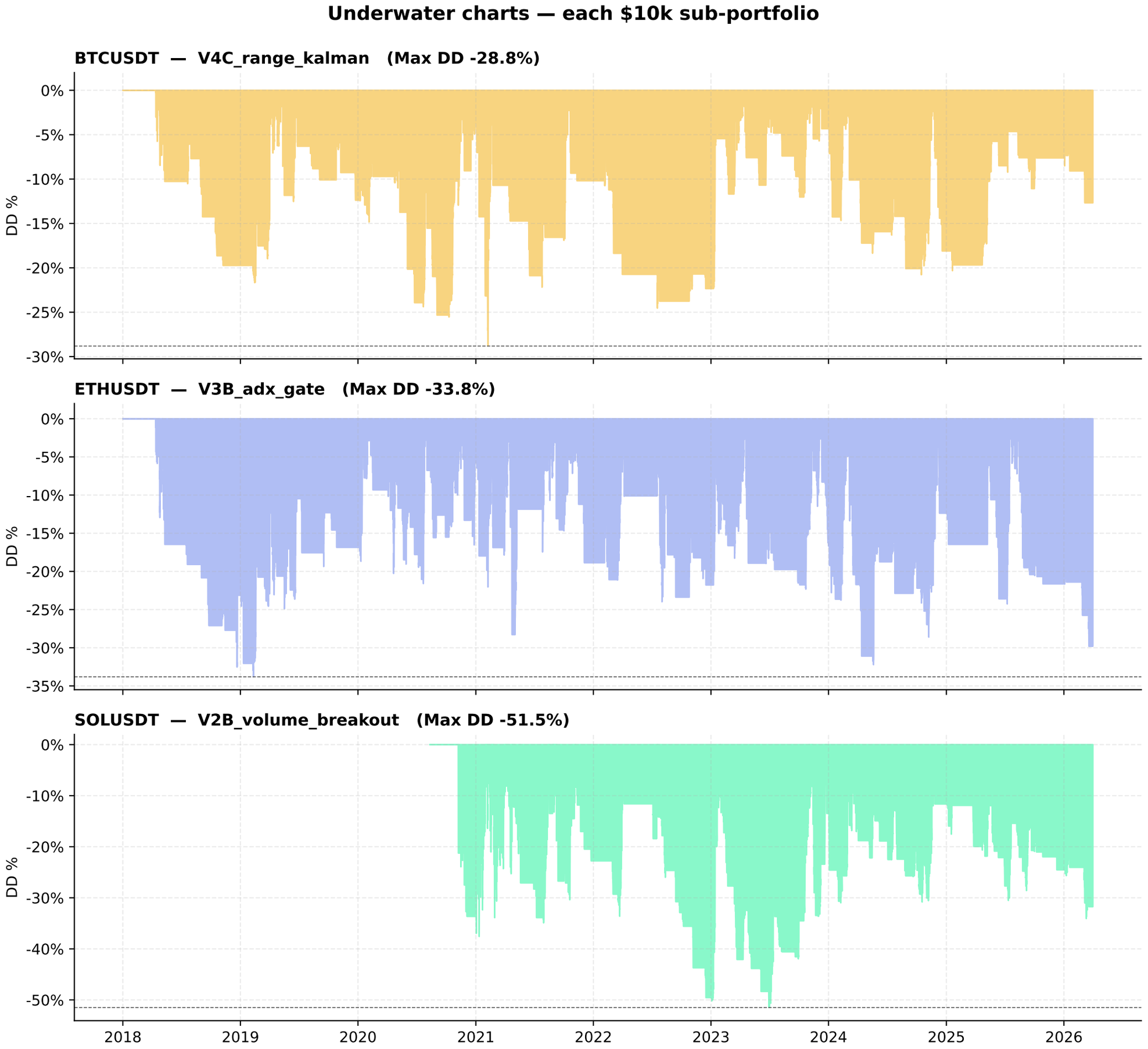
Walk-forward verification (IS 2018-2022 -> OOS 2023-2026, frozen params)

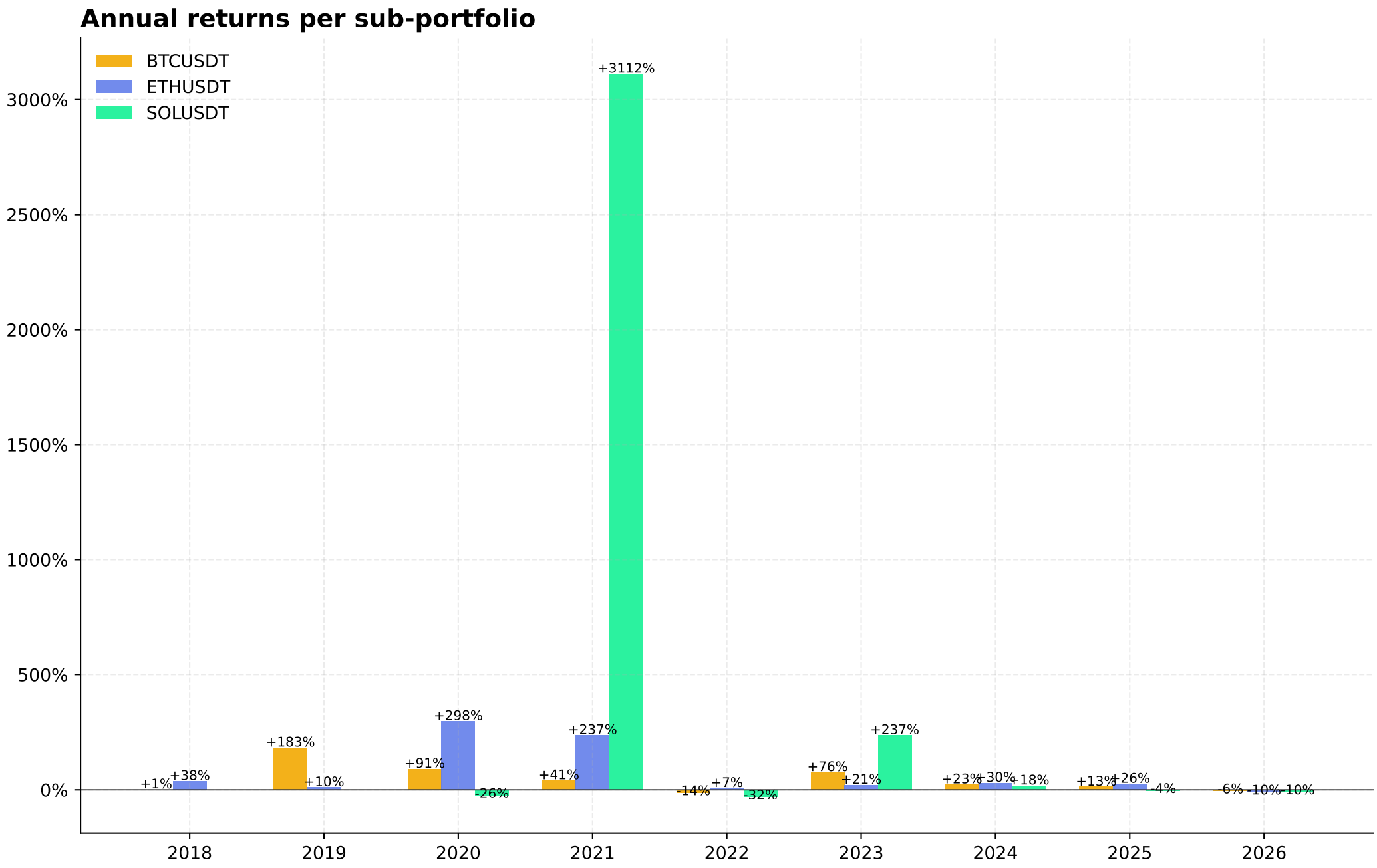
Asset	IS CAGR	IS Sharpe	OOS CAGR	OOS Sharpe	OOS DD
BTCUSDT	+46.5%	1.39	**+22.0%**	**0.97**	-20.8%
ETHUSDT	+85.9%	1.54	**+13.3%**	**0.54**	-32.2%
SOLUSDT	+221.7%	1.72	**+25.6%**	**0.71**	-41.4%

All three strategies passed 5 independent robustness tests (see page 5)

Execution: next-bar OPEN after closed-bar signal | Pine parity enforced


Per-asset equity curves — $10,000 starting each (log scale)
BTCUSDT V4C_range_kalman
ETHUSDT V3B_adx_gate
SOLUSDT V2B_volume_breakout
2018 winter
COVID
BTC ATH1
LUNA/FTX
Halving rally
$1,000,000
$100,000
$10,000
2018
2019
2020
2021
2022
2023
2024
2025
2026


Underwater charts — each $10k sub-portfolio
BTCUSDT — V4C_range_kalman (Max DD -28.8%)
ETHUSDT — V3B_adx_gate (Max DD -33.8%)
SOLUSDT — V2B_volume_breakout (Max DD -51.5%)
DD %
0%
-5%
-10%
-15%
-20%
-25%
-30%
-35%
-40%
-50%
2018
2019
2020
2021
2022
2023
2024
2025
2026


Annual returns per sub-portfolio
BTCUSDT
ETHUSDT
SOLUSDT
3000%
2500%
2000%
1500%
1000%
500%
0%
+1%
+38%
+183%
+10%
+91%
+298%
-26%
+41%
+237%
+3112%
-14%
+7%
-32%
+76%
+21%
+237%
+23%
+30%
+18%
+13%
+26%
-4%
-6%
-10%
-10%
2018
2019
2020
2021
2022
2023
2024
2025
2026

Robustness scorecard — 5 independent overfitting checks

1. Cross-asset generalization — Sharpe on each asset

Does the winner's strategy also produce Sh>0 when applied to the OTHER two assets? (all yes)

Strategy	on BTC	on ETH	on SOL
V4C_range_kalman	Sh 1.32 CAGR +40% (own)	Sh 1.08 CAGR +38%	Sh 1.41 CAGR +107%
V3B_adx_gate	Sh 0.89 CAGR +27%	Sh 1.26 CAGR +56% (own)	Sh 1.04 CAGR +62%
V2B_volume_breakout	Sh 1.01 CAGR +31%	Sh 1.03 CAGR +41%	Sh 1.35 CAGR +105% (own)

2. Random 2-year windows (200 per asset)

Asset	Sharpe median	% windows >0	% windows >0.5
BTCUSDT	1.40	**100.0%**	93.5%
ETHUSDT	1.10	**100.0%**	93.0%
SOLUSDT	1.33	**98.8%**	89.4%

3. 5-fold cross-validation (disjoint 1.5-yr folds)

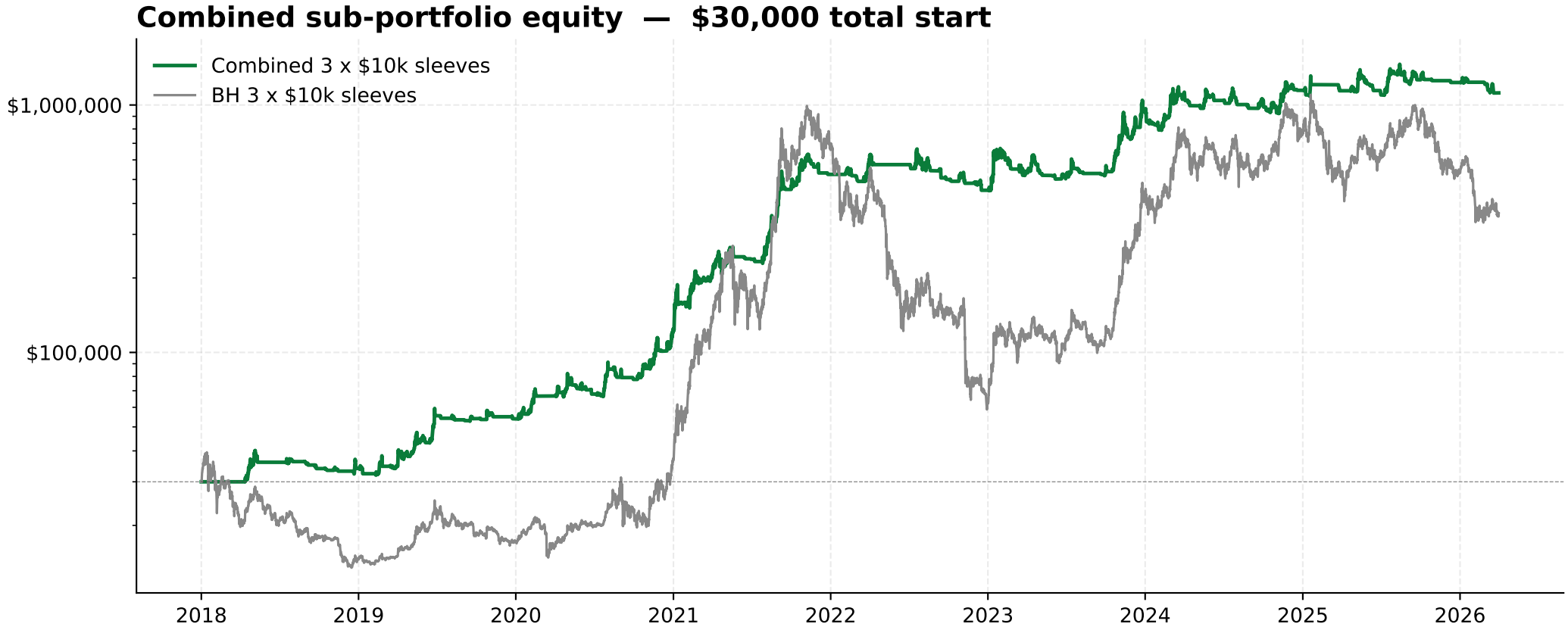
Asset	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
BTCUSDT	**+104%**	**+52%**	**+20%**	**+48%**	**+19%**
ETHUSDT	**+40%**	**+152%**	**+87%**	**+9%**	**+24%**
SOLUSDT	—	**-53%**	**+372%**	**+62%**	**+12%**

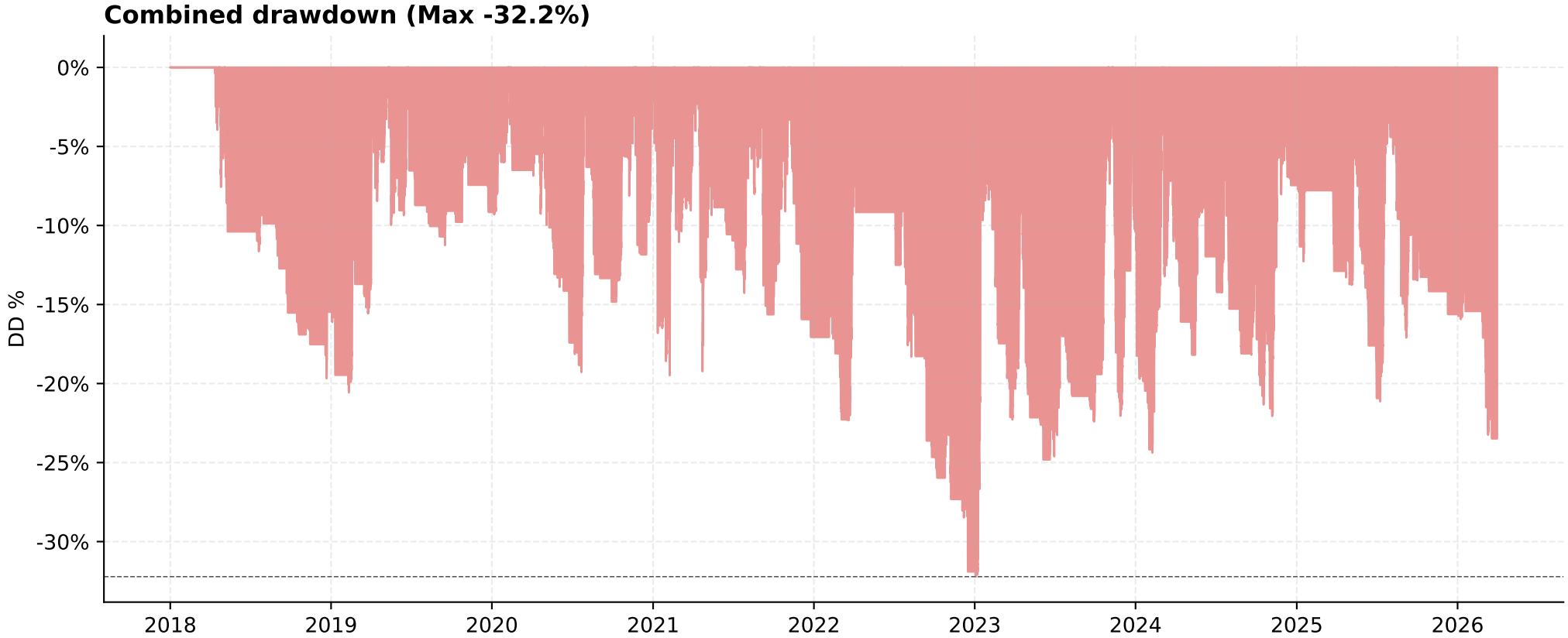
4. Parameter-epsilon grid — every perturbation

Asset	Configs tested	Sharpe range	Calmar range	% profitable
BTCUSDT	81	[0.80, 1.48]	[0.46, 1.65]	**100%**
ETHUSDT	108	[0.97, 1.31]	[0.85, 1.77]	**100%**
SOLUSDT	192	[0.93, 1.39]	[0.77, 2.03]	**100%**

VERDICT — All three strategies pass. None show signs of being curve-fit.

Caveat: SOL fold 2 (2019-2021) had only 3 trades in thin-liquidity era — know your risk.


Combined sub-portfolio equity — $30,000 total start
Combined 3 x $10k sleeves
BH 3 x $10k sleeves
$1,000,000
$100,000
2018
2019
2020
2021
2022
2023
2024
2025
2026


Combined drawdown (Max -32.2%)
DD %
0%
-5%
-10%
-15%
-20%
-25%
-30%
2018
2019
2020
2021
2022
2023
2024
2025
2026

Per-asset strategy logic

BTCUSDT — V4C_range_kalman (4h)

```
Kalman-smoothed baseline (alpha=0.05)
Range band = EMA(|close-kalman|, 100) x 2.5  around baseline
ENTRY: close crosses ABOVE upper band   AND  close > SMA(200)
EXIT: close < lower band  OR  close < SMA(200)
Trailing stop: ATR(14) x 3.5  below highest-since-entry
```

ETHUSDT — V3B_adx_gate (4h)

```
Breakout core: close > highest(high, 30)[1]
           AND volume > SMA(volume, 20) x 1.3
           AND close > SMA(close, 150)
Added filter: ADX(14) > 20 on entry bar
Trailing stop: ATR(14) x 4.5
```

SOLUSDT — V2B_volume_breakout (4h)

```
close > highest(high, 30)[1]
AND volume > SMA(volume, 20) x 1.3
AND close > SMA(close, 150)
Exit: close < SMA(close, 150) regime break
Trailing stop: ATR(14) x 4.5
```

Common execution rules (all assets)

```
- Signal on closed bar i  ->  fill at OPEN of bar i+1 (no look-ahead)
- Commission 0.1 % per side  (Binance spot)
- Slippage 5 ticks
- Position size 100 % of sub-account equity
- No pyramiding, no shorts
```

Honest caveats

```
- ETH IS->OOS Sharpe 1.54 -> 0.54 is the widest gap. Expect live 0.5-0.8.
- SOL fold-2 (2019-07 to 2021-01) had only 3 trades due to thin liquidity.
- All strategies are long-only.  Shorts hurt in every test (crypto bull bias).
- Regime filter (SMA150 / SMA200) sidelines in macro bear markets.
```